

VS Code, PowerShell and Terminal

Development environment, terminal, PATH, variables, venv, pip and useful QA commands.

VS Code

Visual Studio Code is an extensible editor widely used for automation. It supports code editing, extensions, debugging, Git and integrated terminals.

PowerShell, CMD and Git Bash

CMD is the traditional Windows shell. PowerShell combines a shell with an automation language. Git Bash provides a Bash-like environment. The VS Code terminal simply hosts one of these shells.

PATH

PATH is an environment variable that tells the system where to look for executables. If python, git or java is not recognized, installation or PATH may be incorrect.

Python virtual environment

```
python -m venv .venv creates an isolated environment. In PowerShell,  
.\.venv\Scripts\Activate.ps1 activates it. This prevents dependency conflicts between projects.
```

pip and requirements.txt

```
pip installs Python packages. requirements.txt records dependencies so another computer can  
reproduce the environment with pip install -r requirements.txt.
```

Environment variables and secrets

Credentials and tokens should not be hardcoded. Environment variables and CI/CD secrets provide sensitive values at runtime.

Debugging in VS Code

Breakpoints, variable inspection and step-by-step execution help investigate automation. The terminal and debugger solve different problems.

First diagnostics

Use version commands such as `python --version`, `pip --version`, `git --version`, `java -version` and `node --version`. Knowing which executable is used reduces configuration issues.